



HK IMO Training 2019-2020
Problem Set 5 Suggested Solutions



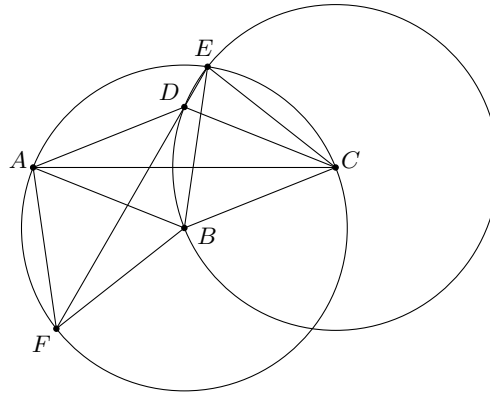
Problem 17. Mandy and Samuel play the following game by making their moves alternately; Mandy moves first. Initially they have a large piece of modelling clay. By a move, Mandy cuts one of the existing pieces into three (of arbitrary sizes), and Samuel merges two existing pieces into one. Mandy wins if at some moment there appears to be 100 pieces of equal weights. Can Samuel prevent Mandy from winning the game?

Solution. No. WLOG assume the weight of the modelling clay is 10^5 . Each time Mandy can choose the heaviest piece and cut two pieces of weight 1 out of it. This can always be done since we will show that the number of moves does not exceed 10^4 .

We claim that the number of pieces of weight 1 is increased by at least 1 after every 100 moves. If this is not the case, that means Samuel merges the two pieces of weight 1 to form a piece of weight 2 each time. After 100 moves, there are 100 pieces of weight 2, which means Mandy has already won. So our claim holds.

Using this claim, there are at least 100 pieces of weight 1 after $100 \times 100 = 10^4$ moves. Thus, Mandy will have won by that time.

Problem 18. The rhombus $ABCD$ is given. Let E be one of the points of intersection of the circles Γ_B and Γ_C , where Γ_B is the circle with centre B and passes through C , and Γ_C is the circle with centre C and passes through B . The line ED intersects Γ_B at point F . Find the value of the angle $\angle AFB$.



Solution. Since C is the circumcentre of $\triangle BDE$, and AC bisects $\angle DCB$, we have $\angle DEB = \angle ACB$. As $BE = BF$, we have $\angle BFE = \angle FEB = \angle ACB$. Therefore,

$$\angle BFA = \angle BFE + \angle EFA = \angle ACB + \angle ECA = \angle ECB.$$

Since $BC = BE = CE$, we have $\angle ECB = 60^\circ$. This implies $\angle ECB = \pm 60^\circ$, so that $\angle BFA = 60^\circ$ or 120° . As $\angle BAF = \angle BFA < 90^\circ$, we must have $\angle BFA = 60^\circ$.

Problem 19. Let p and q be prime numbers. Show that if $p + q^2$ is a perfect square then for any positive integer n the number $p^2 + q^n$ is not a perfect square.

Solution. Let $p + q^2 = a^2$ where a is a positive integer. Then $p = a^2 - q^2 = (a - q)(a + q)$. Since p is a prime and $a + q > 0$, we must have $a + q = p$ and $a - q = 1$. It follows that $a = q + 1$ and $p = a^2 - q^2 = 2q + 1$.

Suppose by contradiction that $p^2 + q^n = b^2$ where b is a positive integer. We have that $q^n = b^2 - p^2 = (b - p)(b + p)$, and therefore $b - p = q^i$ and $b + p = q^j$ where $i < j$. Then

$$2p = (b + p) - (b - p) = q^j - q^i = q^i(q^{j-i} - 1).$$

If $i = 0$, we have that $b - p = 1$ and $b + p = q^n$. Then $q^n = b + p = 2p + 1 = 4q + 3$. This implies $q \mid 3$ and hence $q = 3$. The relation becomes $3^n = 15$, which has no integer solutions.

If $i > 0$, then from $2p = q^i(q^{j-i} - 1)$ we find that $q \mid 2p$, where $2p = 4q + 2$. This implies $q \mid 2$ and therefore $q = 2$ and $p = 2q + 1 = 5$. The equation becomes $10 = 2^i(2^{j-i} - 1)$. It follows that $i = 1$ and that $2^{j-i} - 1 = 5$, which has no integer solutions.

Therefore, $p^2 + q^n$ cannot be a perfect square.

Problem 20. There are n cards on the desk. Each of the n cards has one of the numbers from 1 to n and all numbers on the n cards are distinct. The cards are initially stacked so that the numbers on the cards increase from top to bottom. Let A be an action to move the top card of the stack to the bottom of the stack and B an action to remove the top card of the stack out of the desk. Repeat actions in the following order until the last one is left on the desk: ABBABBABBABB $\dots$.

Let $L(n)$ be the number on the last remaining card. Find all the positive integers k satisfying $L(3k) = k$.

Solution. Consider any positive integer $n \geq 3$ satisfying $L(n) = 1$. In order that the card with the number 1 should be left at the top after the whole stack has been turned once, n must be a multiple of 3. Also, considering the remaining cards, $\frac{n}{3}$ should be 1, 2, or a multiple of 3. In the latter case, we can argue in the same way so that $\frac{n}{3^2}$ should be 1, 2, or a multiple of 3. Repeating the process, we know that $\frac{n}{3^j}$ must be 1 or 2 for some $j \geq 0$. Conversely, all such n satisfy $L(n) = 1$. The necessary and sufficient condition for $L(n) = 1$ is $n = 3^j$ or $n = 2 \cdot 3^j$ for some $j \geq 0$.

Assume that k satisfies $L(3k) = k$. Consider the moment after $k - 1$ actions. At this moment, the top of the stack is the card with the number k and since this card must be left at the end, $k - 1$ should be a multiple of 3. The number of cards remaining at this moment is

$$3k - (k - 1) \times \frac{2}{3} = \frac{7k + 2}{3}.$$

After this moment, the top card must remain by repeating actions until the last one is left on the desk, that is $L\left(\frac{7k + 2}{3}\right) = 1$. From above, we have $\frac{7k + 2}{3} = 3^j$ or $2 \cdot 3^j$ for some $j \geq 0$. This yields $k = \frac{3^{j+1} - 2}{7}$ or $\frac{2 \cdot 3^{j+1} - 2}{7}$. In order that this is an integer, we have $j + 1 \equiv 2 \pmod{6}$ in the former case, and $j + 1 \equiv 0 \pmod{6}$ in the latter case. Thus, the answer is

$$k = \frac{3^{6m-4} - 2}{7}, \frac{2 \cdot 3^{6m} - 2}{7}$$

for some integer $m \geq 1$.